

TIC 309656476

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 28.920445 d
Epoch (t0) 2997.6132 BTJD
Duration 0.1700 d
Depth 0.050 % (495 ppm)
SNR 24.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 14.1 (gate ≥ 7.0)
Odd/even z-score 1.97 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 62 light-curve dip in this sector: 8.5 sigma t0 refined by 0.3975 phase
Difference-image peak SNR 13.80 (gate ≥ 5.0) Compactness 0.87 (gate ≥ 0.50)
Dimming centroid offset 0.66 px (14 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 309656481
1.29 px away, dTmag +5.82 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

